

Notes: Griffin, Kruger, and Mahajan (RFS, 2026)

Is Fraud Contagious? Social Connections and the Looting of COVID Relief Programs

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1 Big picture

- **Question.** Does fraud spread through social connections? More specifically, did social networks help transmit suspicious Paycheck Protection Program (PPP) borrowing during the COVID relief period?
- **Setting.** The paper studies the geography and diffusion of suspicious PPP loans in the United States. The main outcome is suspicious PPP lending at the ZIP-code and county levels, based on loan-level fraud indicators from Griffin, Kruger, and Mahajan (2023).
- **Core challenge.** Fraud is highly geographically clustered, but clustering can arise for many reasons:
 - some areas may have higher underlying fraud propensity;
 - some areas may have more identity theft or cybercriminal activity;
 - some areas may be socially connected to other areas where fraudulent “methods” are circulating.

The key empirical challenge is to separate *social transmission* from common demographics, common local shocks, and homophily.

- **Main idea.** The authors use the Facebook Social Connectedness Index to construct a measure of each ZIP code’s **social proximity to suspicious lending**: the socially weighted average suspicious-lending rate in other ZIP codes. They then ask whether this exposure predicts local PPP fraud, conditional on county fixed effects and rich controls.
- **Main result.** Social proximity to suspicious lending is an extremely strong predictor of local PPP fraud. In the baseline ZIP-code regression, a one standard deviation increase in social proximity is associated with about a **0.86–0.98 standard deviation** increase in suspicious PPP loans per capita, depending on controls.
- **Explanatory-power result.** Social proximity has an incremental adjusted R^2 over **10 times** as large as the strongest demographic variable and larger than all demographic controls combined.
- **Timing result.** Fraud appears to spread over time. ZIP codes socially close to early high-fraud ZIP codes experience much faster growth in suspicious lending. In regressions, social proximity to 2020 fraud predicts 2021 fraud even controlling for the ZIP code’s own 2020 fraud.
- **Mechanism result.** The paper documents social-media groups discussing PPP/EIDL/unemployment relief, matches group members to PPP loans, and finds that groups using fraud-related language have higher suspicious-loan rates. Socially connected areas also share granular loan-application details and use the same FinTech lenders.
- **Policy takeaway.** Fraud prevention cannot rely only on slow ex post prosecution. In a digital environment, fraudulent opportunities and application instructions can diffuse rapidly through social networks. Program administrators need proactive screening, faster detection, and monitoring of emerging fraud channels.

2 Data and measurement (key objects)

2.1 Institutional setting: COVID relief and PPP fraud

- **Pandemic relief scale.** The paper starts from the broader context of COVID relief programs, including PPP, EIDL, and expanded unemployment insurance.
- **PPP size.** PPP distributed roughly **\$793 billion** in forgivable loans through financial intermediaries.
- **Estimated fraud scale.** Griffin, Kruger, and Mahajan (2023) estimate PPP fraud of about **\$64.2 billion** using individual loan indicators and **\$117.3 billion** using broader county-level indicators, about **14.8%** of the program.
- **Other relief programs.** The paper also discusses official and audit-based estimates of potentially fraudulent EIDL and unemployment-insurance payments. Across PPP, EIDL, and UI, the paper reports fraud estimates between roughly **\$300 billion and \$429 billion**.
- **Why PPP is useful empirically.** PPP loans can be mapped to borrower addresses, ZIP codes, counties, lenders, loan amounts, industries, reported jobs, and approval dates. This makes it possible to study both cross-sectional clustering and time-series diffusion.

2.2 PPP loan-level data and suspicious-loan indicators

- **Loan data.** The PPP data contain **11,469,801 loans** originated by **4,809 lenders**, totaling **\$793 billion**.
- **Coverage.** Loans are observed from the start of the PPP on April 3, 2020 through the program’s end on June 30, 2021, for loans not repaid or canceled as of January 2, 2022.
- **Main loan variables.** The data include business name, address, business type, NAICS industry, loan amount, number of employees, approval date, loan draw, and lender.
- **Primary suspicious-loan flags.** The paper uses four main suspicious-loan indicators from Griffin, Kruger, and Mahajan (2023):
 - missing or ineligible registrations for corporations or LLCs;
 - multiple loans at a single residential address;
 - abnormally high implied compensation relative to industry–CBSA averages;
 - large inconsistencies between jobs reported on the PPP application and jobs reported to the EIDL Advance program.
- **Main fraud outcome.** Most regressions use either:

$$\text{PctFlagged}_i = \frac{\text{number of flagged PPP loans in } i}{\text{total PPP loans in } i},$$

or

$$\text{FlaggedPerCapita}_i = \frac{\text{number of flagged PPP loans in } i}{\text{population of } i}.$$

- **Baseline ZIP-code sample.** The main ZIP-code analysis uses **19,753 ZIP codes**, covering over **95%** of the U.S. population. ZIP codes are required to have at least 25 PPP loans.
- **Average suspicious rate.** In the ZIP-code sample, the weighted average percentage of PPP loans flagged is **12.72%**, with standard deviation **9.32 percentage points**. The average flagged-loans-per-capita rate is **0.0069**.

2.3 Social connections data

- **Source.** Social connections come from the Facebook Social Connectedness Index, based on Bailey et al. (2018b, 2020).
- **Pairwise connection measure.** For two ZIP codes i and j ,

$$SCI_{ij} = \frac{\text{Facebook friendship links between } i \text{ and } j}{\text{Facebook users in } i \times \text{Facebook users in } j}.$$

- **Why normalization matters.** Dividing by the product of the number of users in the two ZIP codes makes the measure capture connection *intensity*, rather than mechanically reflecting ZIP codes with more Facebook users.
- **Timing.** The social-connection data are measured in 2021, but the authors argue that such connections are stable over time. Therefore, the measure is treated as capturing general social connections rather than connections created by the PPP itself.
- **Main exposure variable.** Social proximity to suspicious PPP lending is a socially weighted average of suspicious lending in other locations.

2.4 Demographic, economic, cultural, and social-media data

- **Demographics.** ZIP-code and county demographics come from the U.S. Census and IRS Statistics of Income. Controls include population density, average household income, poverty rate, race/ethnicity, college education, and prepandemic unemployment.
- **Crime and corruption proxies.** County-level variables include identity-theft reports, FinCEN suspicious activity reports, public-corruption convictions, religious affiliation, and Ashley Madison usage.
- **Social capital.** ZIP-code social capital measures come from Chetty et al. (2022a, b), including economic connectedness, clustering, support ratio, volunteering rate, and civic organizations.
- **Physical distance.** Distances between ZIP codes and counties come from the NBER and are used to build a physical-proximity analogue of social proximity.
- **Social media.** The authors identify **233** social-media groups on a large platform discussing PPP and related pandemic relief programs. They collect member data from **136 public groups**, with **123,263 unique users**, and match **17,615** users to PPP loans based on unique names.

2.5 Key descriptive facts

- **Geographic concentration.** Some counties have almost no PPP fraud, while others have over **30%** of loans flagged.
- **Within-county variation.** The paper documents large variation even within the same county, with ZIP-code flag rates ranging from about **5%** to above **45%**.
- **City examples.** Cook County (Chicago) has a suspicious-loan rate of **31.7%**, compared with **8.8%** in New York County and **6.1%** in Los Angeles County.
- **FinTech time pattern.** Suspicious lending grew especially rapidly for FinTech lenders. The median county suspicious-loan rate for FinTech loans rose from **6.9%** in April 2020 to **28.7%** by May 2021.
- **Traditional-lender contrast.** Traditional lenders also show some growth in suspicious lending, but the levels are much lower; the median traditional suspicious-loan rate rose from **6.1%** in April 2020 to **10.0%** by May 2021.

3 Models and empirical specifications (all equations + interpretation)

3.1 Three hypotheses

The paper is not a structural model paper. Its “model” is a set of empirical tests designed around three explanations for the geographic concentration and diffusion of PPP fraud.

- **Hypothesis 1: individual fraud propensity.** Fraud is higher where people had a higher preexisting propensity to commit fraud. Under this hypothesis, local demographics, culture, crime, and corruption should explain most of the geographic variation.
- **Hypothesis 2: cybercrime and identity theft.** Fraud is driven by sophisticated cybercriminals exploiting weak program design. Under this hypothesis, suspicious lending should correlate with identity-theft measures, suspicious financial activity reports, and possibly data-breach-type proxies.
- **Hypothesis 3: social transmission.** Fraud spreads through social networks that transmit both the technical “how-to” information and the rationalization that the behavior is acceptable or low risk.

Interpretation. The empirical strategy asks which of these forces best explains where fraud appears and how it grows. The main finding is not that demographics and crime proxies are irrelevant, but that social connections explain much more of the variation and are more closely tied to diffusion over time.

3.2 County-level local-determinants regression

At the county level, the paper estimates:

$$\text{FlaggedPerCapita}_i = \beta \text{Demographics}_i + \gamma \text{CorruptionCrimeProxy}_i + \text{State}_{s(i)} + \varepsilon_i.$$

Here:

- i is a county;
- $s(i)$ is the state containing county i ;
- $\text{FlaggedPerCapita}_i$ is flagged PPP loans divided by population;
- Demographics_i includes income, poverty, unemployment, education, population density, and race;
- $\text{CorruptionCrimeProxy}_i$ includes public corruption, religious affiliation, Ashley Madison usage, identity theft, and FinCEN suspicious activity reports;
- state fixed effects absorb state-level policy, pandemic, or enforcement differences.

Interpretation. This regression asks whether preexisting area characteristics can explain suspicious PPP lending. The paper finds some evidence that they matter, but they do not come close to explaining as much as social proximity.

3.3 ZIP-code demographic and social-capital regression

At the ZIP-code level, the authors estimate:

$$\text{FlaggedPerCapita}_i = \beta \text{Demographics}_i + \gamma \text{SocialCapital}_i + \text{County}_{c(i)} + \varepsilon_i.$$

Here:

- i is a ZIP code;
- $c(i)$ is the county containing ZIP code i ;
- county fixed effects compare ZIP codes *within the same county*;
- all variables are standardized, so coefficients are interpretable in standard deviation units.

Interpretation. This is a stronger comparison than the county-level regression because the authors compare nearby ZIP codes facing similar county-level institutions and local conditions.

3.4 Construction of social proximity to suspicious lending

The central variable is:

$$\text{SocialProximity}_i = \frac{\sum_{\text{County}(j) \neq \text{County}(i)} \text{SCI}_{ij} \times \text{FlaggedPerCapita}_j}{\sum_{\text{County}(j) \neq \text{County}(i)} \text{SCI}_{ij}}.$$

- SCI_{ij} is the strength of Facebook social connections between ZIP codes i and j .
- $\text{FlaggedPerCapita}_j$ is suspicious PPP loans per capita in ZIP code j .

- The baseline measure excludes ZIP codes in the same county as the focal ZIP code.

Interpretation. A ZIP code has high social proximity to fraud if it is strongly connected to other ZIP codes where suspicious PPP lending is high. Excluding same-county ZIP codes helps reduce the concern that the variable is just measuring local geography.

3.5 Baseline social-proximity regression

The main regression is:

$$\text{FlaggedPerCapita}_i = \beta \text{SocialProximityToFraud}_i + \theta \text{Controls}_i + \text{County}_{c(i)} + \varepsilon_i.$$

The controls include demographics, social-geography controls, and FinTech market share. All main variables are standardized.

Interpretation.

- The coefficient β asks: among ZIP codes in the same county, do those more socially connected to high-fraud ZIP codes have higher suspicious PPP lending?
- The answer is yes: the coefficient is large, stable, and statistically significant.
- In the detailed-control specification, $\beta = 0.861$, meaning a one standard deviation increase in social proximity is associated with a 0.861 standard deviation increase in suspicious PPP lending per capita.

3.6 Heterogeneity specification

To test whether the effect is concentrated in specific demographic groups, the authors estimate:

$$\begin{aligned} \text{FlaggedPerCapita}_i = & \beta_{\text{below}} (\text{Proximity}_i \times 1[\text{Demo}_i < \text{DemoMedian}]) \\ & + \beta_{\text{above}} (\text{Proximity}_i \times 1[\text{Demo}_i \geq \text{DemoMedian}]) \\ & + \gamma 1[\text{Demo}_i \geq \text{DemoMedian}] + \theta \text{Controls}_i + \text{County}_{c(i)} + \varepsilon_i. \end{aligned}$$

Interpretation.

- β_{below} is the social-proximity effect for below-median ZIP codes on a given demographic variable.
- β_{above} is the corresponding effect for above-median ZIP codes.
- The estimates are very similar across income, poverty, population density, race, education, unemployment, and technology-use splits.
- This supports a broad social-transmission interpretation rather than an effect concentrated in one demographic group.

3.7 Propagation over time

To test whether fraud spreads over time through social connections, the paper estimates:

$$\text{FlaggedPerCapita}_i^{2021} = \beta \text{SocialProximityToFraud}_i^{2020} + \gamma \text{FlaggedPerCapita}_i^{2020} + \theta \text{Controls}_i + \text{County}_{c(i)} + \varepsilon_i.$$

Interpretation.

- The regression controls for the ZIP code’s own fraud level in 2020.
- Therefore, β measures whether 2021 fraud is predicted by fraud in socially connected regions, above and beyond persistence in the focal ZIP code itself.
- The coefficient is 0.444 using all loans, 0.329 for FinTech loans, and only 0.0341 for traditional loans.
- The time-series evidence is strongest for FinTech loans, consistent with the paper’s broader claim that FinTech platforms and social networks helped amplify suspicious lending.

3.8 Physical proximity and distant social connections

The paper constructs a physical-proximity analogue:

$$\text{PhysicalProximity}_i = \frac{\sum_{\text{County}(j) \neq \text{County}(i)} \frac{1}{\text{Distance}_{ij}} \times \text{FlaggedPerCapita}_j}{\sum_{\text{County}(j) \neq \text{County}(i)} \frac{1}{\text{Distance}_{ij}}}.$$

Then it estimates:

$$\text{FlaggedPerCapita}_i = \beta \text{SocialProximityToFraud}_i + \gamma \text{PhysicalProximityToFraud}_i + \theta \text{Controls}_i + \text{County}_{c(i)} + \varepsilon_i.$$

Interpretation.

- Physical proximity matters when entered alone.
- But once social proximity is included, physical proximity becomes small or negative.
- Social proximity remains large: 1.113 in the joint regression.
- The effect also remains large when social proximity is calculated using only ZIP codes at least 100 or 500 miles away.

3.9 Mechanism regressions: lender choice and loan details

For lender choice, the paper estimates:

$$\text{LoansPerCapitaByLender}_i = \beta \text{SocialProximityToLender}_i + \gamma \text{LoansPerCapita}_i + \theta \text{Controls}_i + \text{County}_{c(i)} + \varepsilon_i.$$

Here, $\text{SocialProximityToLender}_i$ is constructed like social proximity to fraud, but it replaces suspicious loans in other ZIP codes with loans originated by the specific lender or platform.

Interpretation.

- If social networks only capture similar local characteristics, they should not strongly predict highly granular choices such as which FinTech platform borrowers use.
- But social proximity strongly predicts use of FinTech lenders, Womply, and BlueAcorn.
- It does not similarly predict use of Bank of America or JPMorgan Chase.
- This is consistent with social networks transmitting specific application advice and platform recommendations.

4 Identification and instruments (what makes the estimates credible)

4.1 Main identification problem: reflection, homophily, and omitted variables

- The biggest concern is the **reflection problem**: similar fraud rates across socially connected ZIP codes may reflect social transmission, but may also reflect shared characteristics or common shocks.
- For example, if poorer ZIP codes both have more fraud and are more connected to each other, social proximity could proxy for poverty rather than information diffusion.
- If fraud is higher in areas with similar culture, race, crime, or identity-theft exposure, social proximity may pick up homophily rather than contagion.
- The authors do not claim to solve the reflection problem with one clean instrument. Instead, they build a cumulative case using many complementary tests.

4.2 County fixed effects and rich controls

- The baseline ZIP-code specifications include **county fixed effects**, so identification comes from comparing ZIP codes inside the same county.
- This absorbs county-level policy, enforcement, pandemic, lender, and economic shocks.
- The controls include detailed local demographics, local social-geography variables, and FinTech market share.
- The social-proximity coefficient changes little when controls are added, which the authors interpret through the logic of Oster (2019): unobservables would need to be much more important than observables to explain away the result.

4.3 Social proximity versus physical proximity

- Physical proximity is an important alternative mechanism because fraud could spread through local advertising, brokers, or nearby addresses.
- The paper constructs physical proximity to fraud using inverse geographic distance.

- Social proximity remains strongly positive after controlling for physical proximity.
- Physical proximity does not remain positive after controlling for social proximity.
- The result also holds when the social-proximity measure uses only distant ZIP codes at least **100** or **500** miles away.

4.4 Timing evidence

- The time dimension is useful because social transmission predicts that early high-fraud areas should affect later fraud in socially connected areas.
- The authors identify initial high-fraud ZIP codes in the first month of PPP and track other ZIP codes based on their social connectedness to those initial hotspots.
- ZIP codes with high social exposure to initial hotspots experience much faster fraud growth, even when initially matched on similar fraud rates.
- In regressions, 2020 fraud in socially connected areas predicts 2021 fraud in the focal ZIP code, even controlling for the focal ZIP code’s own 2020 fraud.

4.5 Granular mechanism evidence

- The strongest evidence against a pure homophily story is that social connections predict specific application decisions:
 - identical loan features;
 - industry–amount–reported–employees clusters;
 - the choice of specific FinTech lenders and platforms.
- Social-media group evidence also links individual-level group membership to suspicious loans.
- These patterns are harder to explain with broad omitted variables, because broad omitted variables should not naturally generate the same lender choice or identical loan-feature clusters across socially connected regions.

4.6 Limitations

- The paper cannot fully prove individual-level causal peer effects for every borrower.
- Facebook connections proxy for broader social connections, but they do not reveal the exact information path between two borrowers.
- Social-media matching uses unique names and is validated manually on a subsample, but name matching can still generate some false positives or miss many true matches.
- Suspicious-loan flags are not legal determinations of fraud. The paper uses validated indicators of suspicious activity, not criminal convictions.

- The strongest interpretation is therefore: the evidence is highly consistent with social transmission and difficult to reconcile with simple homophily alone.

4.7 Relation to the literature

- **Fraud and crime diffusion.** The paper connects to theories of social interaction in crime, including social norms, cascades, and differential association.
- **Social finance.** It builds on work showing that social networks affect stock-market participation, housing beliefs, refinancing, mortgage default, and bank choice.
- **COVID relief programs.** It complements work studying PPP allocation, FinTech lenders, and the labor-market effects of relief programs by focusing on fraud diffusion.
- **Policy design.** The paper speaks to program administration: rapid disbursement with weak screening can interact with social networks to create large fraud externalities.

5 Tables and figures

5.1 Table 1: summary statistics

Read:

- The ZIP-code sample has **19,753** observations and the county sample has **3,187** observations.
- At the ZIP-code level, the average share of PPP loans flagged is **12.72%**; the standard deviation is **9.32 percentage points**. This already shows large cross-sectional dispersion.
- The average number of PPP loans per capita is **0.0503**; the average flagged loans per capita is **0.0069**.
- At the county level, the average flagged share is **12.32%**; the broader “excess share” measure averages **22.0%**.
- The table also defines the key social and physical proximity measures, which are used throughout the empirical analysis.

5.2 Figure 1: geography and time path of suspicious lending

Read:

- Panel A shows that suspicious lending is geographically concentrated. High-fraud areas appear around New Orleans, Atlanta, parts of Louisiana, Mississippi, Georgia, Chicago, South Carolina, and scattered urban or regional pockets elsewhere.
- Panel B shows large within-county variation. Even inside the same county, ZIP codes can have flag rates ranging from about **5%** to more than **35%**.
- Panel C shows that suspicious lending grows over time, especially among FinTech lenders.

Table 1
Summary statistics

A. ZIP code level						
	Mean	SD	p25	p50	p75	N
Pct. PPP Flagged	0.1272	0.0932	0.0685	0.0969	0.153	19,753
PPP Loans Per Capita	0.0503	0.0562	0.0273	0.0384	0.0564	19,753
Flagged PPP Per Capita	0.0069	0.0101	0.0023	0.0037	0.0071	19,753
Social Proximity to Suspicious PPP Lending	0.0047	0.0021	0.0034	0.0042	0.0054	19,753
Physical Proximity to Suspicious PPP Lending	0.0039	0.0008	0.0034	0.0038	0.0042	19,753
B. County level						
	Mean	SD	p25	p50	p75	N
Pct. PPP Flagged	0.1232	0.0726	0.0704	0.0982	0.1567	3,187
PPP Loans Per Capita	0.0422	0.0221	0.0290	0.0368	0.0464	3,187
Flagged PPP Per Capita	0.0055	0.0053	0.0024	0.0035	0.0064	3,187
Excess Share	0.2200	0.1343	0.1143	0.1852	0.3077	3,187
Excess Share Per Capita	0.0054	0.0055	0.0021	0.0035	0.0067	3,187
Social Proximity to Suspicious PPP Lending	0.0037	0.0008	0.0032	0.0036	0.0040	3,187
Physical Proximity to Suspicious PPP Lending	0.0033	0.0003	0.0030	0.0033	0.0035	3,187

This table provides summary statistics. Panel A presents summary statistics for variables at the ZIP code level, and panel B presents them at the county level. *Pct. PPP Flagged* represents the percentage of PPP loans flagged by at least one of the four primary flags developed by Griffin, Kruger, and Mahajan (2023). *PPP Loans Per Capita* is the ratio of PPP loans to the population of the ZIP code or county. *Flagged PPP Per Capita* is the ratio of PPP loans flagged by at least one of the four primary flags developed by Griffin, Kruger, and Mahajan (2023) to the population of the ZIP code or county. *Social Proximity to Suspicious PPP Lending* is the weighted average *Flagged PPP Per Capita* in each ZIP code or county's socially proximate ZIP codes or counties, with weights determined by the Social Connectedness Index (from Bailey et al. (2018b, 2020)) between pairs of ZIP codes or counties. *Physical Proximity to Suspicious PPP Lending* is the weighted average *Flagged PPP Per Capita* in each ZIP code or county's physically proximate ZIP codes or counties, with weights based on inverse distances between pairs of ZIP codes or counties. Social and physical proximity measures at the ZIP code (county) level exclude areas within the focal ZIP code's (county's) county (state). *Excess Share* is the share of first-draw business loans that exceed Census establishment counts at the industry-county pair level. *Excess Share Per Capita* is calculated as the ratio of first-draw business loans exceeding Census establishment counts at the industry-county pair level to county population. All statistics are weighted by the number of PPP loans in each region.

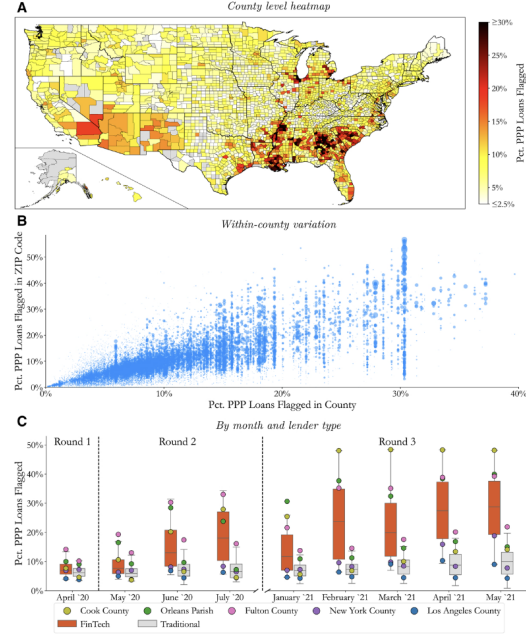


Figure 1
Geography of PPP flagged loans
This figure illustrates geographic variation in the percentage of flagged PPP loans. Panel A displays the percentage of flagged loans in each county. Panel B highlights within-county variation. Panel C presents the distributions of flag rates across counties and over time for loans originated by FinTech and traditional lenders. In panel A, counties are colored according to the color bar to the right of the map, with counties having fewer than 100 loans shaded in gray. In panel B, the vertical axis represents the percentage of flagged loans in each ZIP code, while the horizontal axis shows the percentage of flagged loans in the corresponding county. The size of each dot corresponds to the number of loans in the ZIP code, and only ZIP codes with at least 25 loans are included. In panel C, boxplots are weighted by the number of loans in each county-lender type-month cell, with cells containing fewer than 25 loans excluded. Additionally, flag rates for five specific counties are displayed.

- The contrast between FinTech and traditional lenders is central: by May 2021, the median county flag rate is about **28.7%** for FinTech loans but only about **10.0%** for traditional loans.

5.3 Table 2: county-level determinants of fraud

Read:

- Fraud is higher in counties with higher poverty rates and higher nonwhite population shares.
- Identity theft is one of the strongest county-level predictors: the historical identity-theft coefficient is about **0.340**, and the contemporaneous identity-theft coefficient is about **0.392**.
- Religious affiliation and public corruption are positively related to suspicious lending, though the public-corruption relation is weaker.
- The table supports the idea that local traits and crime-related proxies matter, but these variables are not the paper's main explanatory force.

5.4 Table 3: ZIP-code demographics and social capital

Read:

- At the ZIP-code level, suspicious lending is higher in less densely populated areas, higher-income areas, and ZIP codes with larger nonwhite population shares.

Table 2
County-level determinants of fraud: Demographics, corruption, and crime
Dep. Variable: Flagged by at Least One Primary Measure Per Capita in County

	(1)	(2)	(3)
Log Population Density	0.0402 (0.32)	-0.0876 (-0.52)	-0.0785 (-0.59)
Log Average Income	0.371** (2.55)	0.0305 (0.21)	0.0535 (0.59)
Poverty Rate	0.241** (2.25)	0.183*** (3.60)	0.180*** (3.76)
Prepandemic Unemployment	0.0293 (0.34)	0.0173 (0.42)	0.000359 (0.01)
Pct. College Educated	-0.137 (-1.14)	-0.00512 (-0.08)	0.0122 (0.24)
Pct. Nonwhite	0.433*** (3.38)	0.335*** (2.85)	0.297** (2.37)
Public Corruption		0.0908* (1.80)	0.109* (1.91)
Religious Affiliation		0.101*** (3.63)	0.0946*** (3.39)
Ashley Madison Usage		0.0941 (0.84)	0.0537 (0.49)
Identity Theft		0.340*** (3.91)	
Identity Theft, Contemporaneous			0.392*** (4.97)
FinCEN SARs		0.0564 (1.13)	
FinCEN SARs, Contemporaneous			0.113** (2.49)
State FE	Yes	Yes	Yes
Observations	1,557	1,557	1,557
Adjusted R^2	0.719	0.768	0.782
Adjusted Within R^2	0.382	0.488	0.519

This table examines the county-level relationship between the number of flagged loans per capita and various measures of demographics, corruption, and crime. *Identity Theft* refers to the average number of identity theft reports to the FTC per capita in each CBSA from 2015 to 2019. *FinCEN SARs* represents the average number of suspicious activity reports per capita in each county from 2015 to 2019 (winsorized at the 95th percentile). *Public Corruption* is defined as the number of public corruption convictions per million residents in each judicial district from 2010 to 2019 (as reported by the Department of Justice). *Religious Affiliation* measures the percentage of a county's population with a religious affiliation as of 2020 (as reported by the Association of Religious Data Archives). *Ashley Madison* indicates the paid Ashley Madison usage rate in each county (as reported by Griffin, Kruger, and Maturana (2019)). Column (3) uses versions of the *Identity Theft* and *FinCEN SARs* variables based on data from 2020 and 2021. To ensure nationally representative estimates, we use weighted least squares regressions, with weights corresponding to the number of loans in each county. All variables (both independent and dependent) are standardized to have a mean of 0 and a standard deviation of 1. Fixed effects are specified at the bottom of each column. t -statistics based on robust standard errors are clustered at the state level in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 3
ZIP code-level determinants of PPP fraud: Demographics and social capital
Dep. Variable: Flagged by at Least One Primary Measure Per Capita in ZIP code

	(1)	(2)	(3)
Log Population Density	-0.141*** (-3.15)	-0.0735*** (-2.82)	-0.216*** (-3.29)
Log Average Income	0.224*** (2.59)	0.216*** (2.71)	0.149* (1.90)
Poverty Rate	0.0420 (0.65)	0.0621 (1.15)	-0.0171 (-0.37)
Prepandemic Unemployment	0.263 (1.56)	0.156 (1.27)	0.280* (1.77)
Pct. College Educated	0.0119 (0.21)	-0.00372 (-0.06)	0.0547 (0.81)
Pct. Nonwhite	0.466*** (4.16)		0.427*** (4.46)
Pct. Black		0.572*** (4.56)	
Pct. Asian		0.0130 (0.36)	
Pct. Native		-0.0505** (-2.17)	
Pct. Other Race		-0.0516 (-1.15)	
Pct. Hispanic/Latino		0.0287 (0.42)	
Economic Connectedness			0.0125 (0.22)
Clustering			-0.129 (-1.20)
Support Ratio			-0.153*** (-3.69)
Volunteering Rate			-0.131** (-2.45)
Civic Organizations			0.0943*** (3.45)
County FE	Yes	Yes	Yes
Observations	19,753	19,753	17,301
Num. Counties	2,338	2,338	2,141
Adjusted R^2	0.603	0.650	0.635
Adjusted Within R^2	0.313	0.394	0.373

This table examines the ZIP code-level relationship between the number of flagged loans per capita and various demographic factors and measures of social capital. The demographic variables are sourced from the American Community Survey (2015–2019), while the social capital measures are obtained from Chetty et al. (2022a, b). To ensure nationally representative estimates, we use weighted least squares regressions, with weights based on the number of PPP loans in each ZIP code. All variables (both independent and dependent) are standardized at the ZIP code level to have a mean of 0 and a standard deviation of 1. ZIP codes are filtered to include only those with at least 25 loans. Fixed effects are specified at the bottom of each column. t -statistics based on robust standard errors are clustered by county in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

- In the detailed race specification, the percentage Black coefficient is large and positive, while the percentage Native American coefficient is negative.
- Social-capital measures matter in intuitive ways: support ratio and volunteering are negatively related to fraud, while civic organizations are positively related.
- The positive civic-organization result is less straightforward; one interpretation is that some civic or local organization networks may also transmit information, but the paper does not lean heavily on that mechanism.

5.5 Figure 2: network of fraud

Read:

- The figure depicts social connections among three high-fraud combined statistical areas: Chicago, Atlanta, and New Orleans.
- Nodes are ZIP codes, node colors represent PPP fraud rates, and edge thickness represents social connection strength.
- High-fraud ZIP codes are disproportionately connected to other high-fraud ZIP codes across cities.
- This figure provides the visual motivation for constructing the social-proximity measure formally.

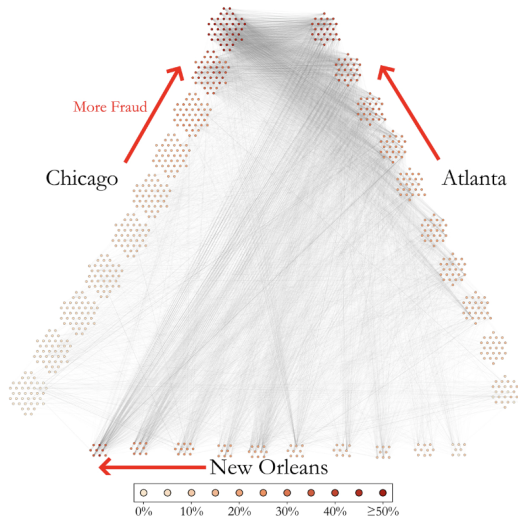


Figure 2
Network of fraud
This figure depicts the network of social connections among three high-fraud CSAs: Chicago, Atlanta, and New Orleans. Chicago ZIP codes are positioned on the left, Atlanta on the right, and New Orleans at the bottom. Each node represents a ZIP code, while the edges indicate connections between pairs of ZIP codes. Connections within the same CSA are excluded. The color of each node reflects the percentage of PPP loans flagged for fraud. The width of each edge corresponds to the strength of social connections between ZIP codes, as measured by the Social Connectedness Index developed by Bailey et al. (2018b, 2020). ZIP codes are divided into 10 groups within each CSA based on their fraud levels. Red arrows point toward higher-fraud nodes within each CSA.

Table 4
Social proximity to suspicious lending
A. Social Proximity

Dep. Variable: Flagged by at Least One Primary Measure Per Capita in ZIP code	(1)	(2)	(3)
Social Proximity to Suspicious Lending	1.054*** (4.01)	0.982*** (5.09)	0.861*** (5.27)
Controls	No	Yes	Yes
Detailed Race Controls	No	No	Yes
County FE	Yes	Yes	Yes
Observations	19,753	19,753	19,753
Num. Counties	2,338	2,338	2,338
R^2	0.699	0.729	0.739
Adjusted R^2	0.659	0.693	0.704
Adjusted Within R^2	0.409	0.469	0.487

B. Incremental R^2 based on panel A, column (3)

Variable	Incremental Adjusted R^2	Incremental Adjusted Within R^2
Social Proximity to Suspicious Lending	0.05261	0.09101
Pct. Black	0.00525	0.00908
Prepandemic Unemployment	0.00525	0.00908
Log Average Income	0.00443	0.00767
Pct. FinTech	0.00275	0.00476
Log Population Density	0.00153	0.00264
Pct. College Educated	0.00122	0.00210
Share Friends Within 150 Miles	0.00114	0.00197
Pct. Other Race	0.00101	0.00176
Share Friends Within 50 Miles	0.00093	0.00161
Poverty Rate	0.00030	0.00052
Pct. Native	0.00025	0.00043
Pct. Hispanic/Latino	0.00005	0.00008
Pct. Asian	-0.00001	-0.00002
Adding All to Social Proximity	0.04503	0.07791

This table examines the ZIP code-level relationship between social proximity to fraud and the number of flagged loans per capita. Panel A presents the relationship between social proximity and fraud rates. Panel B reports the incremental R^2 of each explanatory variable based on the specification shown in column (3) of panel A. Both the incremental adjusted R^2 and the incremental adjusted within-county R^2 are included. The "Adding All to Social Proximity" row shows the incremental adjusted R^2 when all listed control variables are added to a regression that already includes social proximity as a control. The baseline controls include the log of population density, percentage nonwhite, log of average household income, poverty rate, prepandemic unemployment rate, percentage with a college education, the shares of Facebook users in the ZIP code with friends living within 50 and 150 miles, and the FinTech market share of PPP loans in the ZIP code. The detailed race controls replace the percentage nonwhite variable with percentages for Black, Asian, Native American, other races, and Hispanic/Latino populations. To ensure nationally representative estimates, we use weighted least squares regressions, with weights based on the number of PPP loans in each ZIP code. All variables (both independent and dependent) are standardized at the ZIP code level to have a mean of 0 and a standard deviation of 1. ZIP codes are filtered to include only those with at least 25 loans. Fixed effects and controls are specified at the bottom of each column. t -statistics based on robust standard errors are clustered by county in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.6 Table 4: social proximity to suspicious lending

Read:

- This is the central table of the paper.
- Without controls, the coefficient on social proximity is **1.054**; with baseline controls, it is **0.982**; with detailed race controls, it is **0.861**. All are highly statistically significant.
- Panel B shows why the effect is economically important: social proximity alone has an incremental adjusted R^2 of **0.0526** and an incremental within-county adjusted R^2 of **0.0910**.
- The next strongest variables have incremental adjusted R^2 around **0.00525**. Thus, social proximity explains roughly an order of magnitude more variation than any single demographic predictor.
- Adding all controls to a regression already containing social proximity contributes less incremental explanatory power than adding social proximity itself.

5.7 Figure 3: heterogeneity in the social-proximity effect

Read:

- The figure asks whether social transmission is concentrated in specific demographic groups.
- The answer is basically no: the effect is similar above and below median income, poverty, density, nonwhite share, education, and unemployment.

- For example, the effect is about **0.93** in below-median nonwhite-share ZIP codes and about **0.98** in above-median nonwhite-share ZIP codes.
- This broad-based pattern helps address the concern that the social-proximity result is driven by one demographic subgroup.

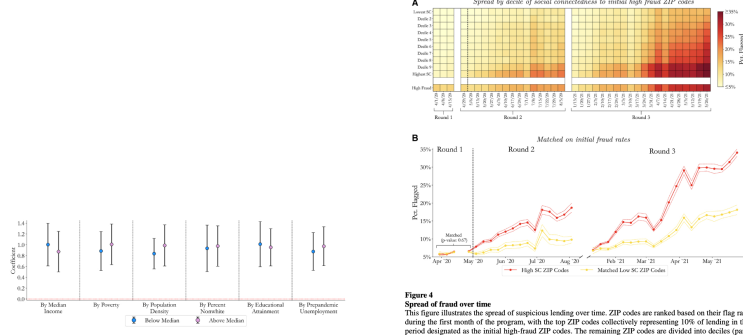


Table 5
Spread across rounds
Dep. Variable: 2021 Flagged by at Least One Primary Measure Per Capita in ZIP code

	(1) All Loans	(2) FinTech Loans	(3) Traditional Loans
Social Proximity to 2020 Suspicious Lending	0.444*** (2.81)	0.329*** (2.70)	0.0341*** (2.45)
2020 Flagged Per Capita	0.473*** (4.21)	0.668*** (3.86)	1.010*** (20.21)
County FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Observations	19,393	19,393	19,393
Num. Counties	2,308	2,308	2,308
Adjusted R ²	0.746	0.816	0.876
Adjusted Within R ²	0.551	0.666	0.868

***This table examines the spread of fraud across rounds. In column (1), social proximity to suspicious lending and the fraud rate are calculated based on all loans. In columns (2) and (3), these measures are calculated based solely on FinTech and traditional loans, respectively. The controls include the log of population density, percentage nonwhite, log of average household income, poverty rate, percentage unemployment rate, percentage with a college education, and the share of Facebook users in the ZIP code with friends living within 50 and 150 miles. To ensure nationally representative estimates, we use weighted least squares regressions, with weights based on the number of PPP loans in each ZIP code. All variables (both independent and dependent) are standardized to have a mean of 0 and a standard deviation of 1. ZIP codes are filtered to include only those with at least 25 loans. Fixed effects and controls are specified at the bottom of each column. z-statistics based on robust standard errors are clustered by county in parentheses. *p < .1, **p < .05, ***p < .01.

5.8 Figure 4 and Table 5: spread across time and PPP rounds

Read:

- Figure 4 tracks ZIP codes based on their social proximity to initial high-fraud ZIP codes.
- Initial high-fraud ZIP codes begin with high fraud and grow from about **13%** during the first month to over **30%** near the end of the program.
- Other ZIP codes start below **7%**, but those most socially connected to initial hotspots experience the fastest growth.
- In the matched analysis, high- and low-social-connection ZIP codes start with nearly identical fraud rates. By the end of the program, the top social-connection group reaches about **35%** flagged loans, while the matched low-connection group remains below **20%**.
- Table 5 makes the timing test regression-based: social proximity to 2020 suspicious lending predicts 2021 suspicious lending, even controlling for the ZIP code's own 2020 fraud.
- The effect is mainly driven by FinTech loans: coefficient **0.329** for FinTech versus only **0.0341** for traditional loans.

5.9 Figure 5 and Table 6: social versus physical proximity

Read:

- Figure 5 shows that social and physical proximity are correlated but distinct.
- Conditional on physical proximity, fraud rates rise strongly with social proximity.

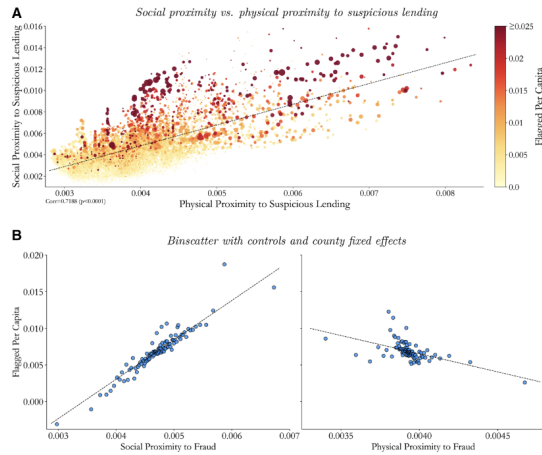


Figure 5
Social and physical proximity to suspicious lending
This figure examines the relationship between social and physical proximity to suspicious lending and flagged loans per capita across ZIP codes. In panel A, each dot represents a ZIP code, is sized according to the number of loans in the ZIP code, and is colored based on the ratio of flagged loans in the ZIP code to its population, as indicated by the color bar. Panel B presents binscatters using ZIP code-level data. Both subpanels in panel B include county fixed effects and control for log population density, percentage nonwhite, log average household income, poverty rate, pre-pandemic unemployment rate, percentage with college education, the share of Facebook friends within 50 and 150 miles, and the FinTech market share of PPP loans in the ZIP code. The left (right) subpanel additionally controls for physical proximity (social proximity) to suspicious lending. To ensure nationally representative estimates, all panels use weighted least squares regressions, with weights based on the number of PPP loans in each ZIP code. ZIP codes with fewer than 25 loans are excluded from the analysis. Dashed lines indicate linear fits.

Table 6
Controlling for and excluding nearby regions
Dep. Variable: Flagged by at Least One Primary Flag Per Capita in ZIP code

	(1)	(2)	(3)	(4)	(5)	(6)
SP Based on:		Baseline		Baseline		
PP Based on:	Baseline		25 Mi Triangular Kernel		≥ 100 Mi	≥ 500 Mi
Social Proximity to Suspicious Lending		1.113*** (4.95)		0.991*** (5.00)	0.615*** (4.95)	0.553*** (3.54)
Physical Proximity to Suspicious Lending	0.382** (2.52)	-0.384*** (-2.81)	0.415*** (3.05)	-0.0190 (-0.29)		
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	19,753	19,753	18,834	18,834	19,753	19,753
Num. Counties	2,338	2,338	2,245	2,245	2,338	2,338
Adjusted R^2	0.618	0.698	0.638	0.695	0.645	0.644
Adjusted Within R^2	0.340	0.477	0.373	0.473	0.385	0.384

This table examines the robustness of the relationship between social proximity to suspicious lending and fraud rates, accounting for fraud near the focal ZIP code. In columns (1) and (2), we present the relationship between the baseline version of physical proximity and fraud rates, both independently and jointly with social proximity. Columns (3) and (4) replace the baseline version of physical proximity with a version based on a 25-mile triangular kernel. The 25-mile triangular kernel assigns a weight of $25-x$ to ZIP codes within 25 miles of the focal ZIP code, where x is the distance to the ZIP code, and a weight of zero to ZIP codes further than 25 miles. Columns (5) and (6) replace the baseline version of social proximity with versions based on ZIP codes that are at least 100 and 500 miles away from the focal ZIP code, respectively. The controls include the log of population density, percentage nonwhite, log of average household income, poverty rate, pre-pandemic unemployment rate, percentage with a college education, and the shares of Facebook users in the ZIP code with friends living within 50 and 150 miles. To ensure nationally representative estimates, we use weighted least squares regressions, with weights based on the number of PPP loans in each ZIP code. All variables (both independent and dependent) are standardized to have a mean of 0 and a standard deviation of 1. ZIP codes are filtered to include only those with at least 25 loans. Fixed effects and controls are specified at the bottom of each column. t -statistics based on robust standard errors are clustered by county in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

- Conditional on social proximity, physical proximity has little positive explanatory power.
- Table 6 confirms this formally: physical proximity alone has a positive coefficient, but when both variables enter, physical proximity becomes negative while social proximity remains large at **1.113**.
- The social-proximity coefficient remains large when calculated using only ZIP codes at least **100 miles** or **500 miles** away. This supports a long-distance information-transmission interpretation.

5.10 Figure 6: social media groups

Read:

- The authors identify social-media groups discussing PPP and related relief programs.
- Some groups appear to provide legitimate guidance, while others use language associated with fraudulent opportunities and application “methods.”
- Among 136 public groups, the authors match **17,615** members to PPP loans. These matched users received **21,321** PPP loans totaling **\$336 million**; **76.2%** were originated by FinTech lenders.
- Groups with higher member suspicious-loan rates use more fraud-associated terms and more often discuss Womply and BlueAcorn.
- The figure gives direct mechanism evidence: fraudulent knowledge and normalization appear to be circulating in public online groups.

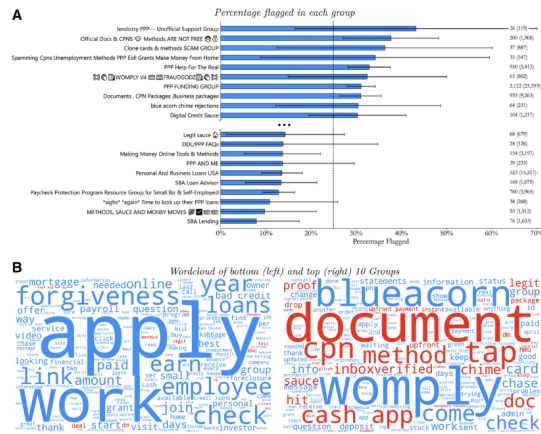


Figure 6
Social media groups
This figure analyzes social media groups on a large platform that discuss the PPP and other pandemic relief programs. Panel A displays the percentage of PPP loans received for members of each group that are flagged as suspicious. Groups with fewer than 25 loans matched to their members are excluded. For clarity, only the 10 groups with the highest and lowest flag rates among loans matched to their members are shown; Figure 1A.12 provides a version of panel A that includes all 72 groups. The number of members in each group matched to a PPP loan, along with the total number of members in each group (in parentheses), is displayed to the right of each bar. Error bars represent 95% confidence intervals adjusted using Bonferroni correction for multiple testing. The dashed line indicates the average flag rate across all loans matched to members of any group. Panel B highlights the 200 most common words used in posts from the bottom (left) and top (right) 10 groups, ranked by the percentage of flagged PPP loans among their members. Common words, also known as stop words (e.g., “the,” “he,” “she,” and “it”), are excluded, along with names of programs and other frequently used terms in pandemic relief contexts (e.g., “PPP,” “EIDL,” “unemployment,” “borrower,” and “lender”). The size of each word corresponds to its frequency within each set of groups. Words displayed in red are primarily associated with nefarious purposes. For five larger groups, the complete universe of posts is unavailable, limiting data collection to approximately 1,700 posts per group.

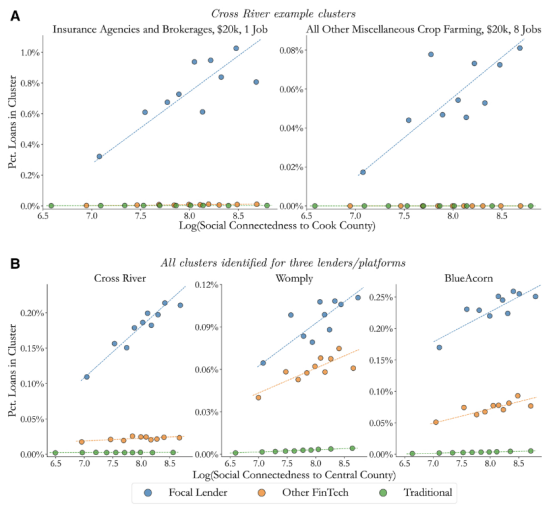


Figure 7
Clustering along loan features
This figure explores clustering along loan features and examines whether these clusters are concentrated in socially connected counties. Panel A provides two specific examples of loan clusters highlighted in Griffin, Kruger, and Mahajan (2023), while panel B presents the average effect across all clusters identified for Cross River, Womply, and BlueAcorn. Each cluster is defined by a combination of three loan features: the industry of the business receiving the loan, the loan amount, and the number of jobs reported. Both panels analyze the relationship between the percentage of a lender’s loans in each county that match all three loan features and the social connectedness of that county to the central county of each cluster. Binscatters are used to visualize these relationships. Each binscatter is run separately, weighted by the total number of loans by the lender in each county, and includes state fixed effects. In panel A, two clusters of loans originated by Cross River are considered. For both clusters, Cook County (Chicago) serves as the central county because it received the majority of loans matching these clusters. In panel B, clusters are identified by combinations of industry, loan amount, and jobs reported that include at least 100 loans originated by a single lender/platform in a single county. The central county for each cluster is defined as the county where the lender/platform has issued the most loans matching that combination of features. Industry is categorized using four-digit NAICS codes, and loan amounts are rounded to the nearest \$100 to account for minor differences. Loans at the maximum allowed per employee are excluded to avoid mechanical clustering at this threshold. The binscatters in panel B are generated similarly to those in panel A, with averages calculated across all clusters identified for each lender/platform. In both panels, blue dots/lines represent loans originated by the focal lender, orange represents loans originated by other FinTech lenders, and green represents loans originated by traditional lenders.

5.11 Figure 7: clustering along loan features

Read:

- Figure 7 examines whether socially connected counties share identical loan features.
- The first Cook County example is **4,299** Cross River loans for **\$20,000** in “Insurance Agencies and Brokerages,” usually with exactly one employee.
- The second Cook County example is **938** Cross River loans for **\$20,000** in “All Other Miscellaneous Crop Farming,” usually with exactly eight employees.
- Counties more socially connected to Cook County have more loans matching these specific feature clusters.
- The generalized cluster analysis finds **40** Cross River clusters, **40** Womply clusters, and **21** BlueAcorn clusters. Traditional lenders have no comparable clusters meeting the same criteria.
- This is strong mechanism evidence because it links social connectedness to precise application details, not just broad fraud rates.

5.12 Table 7: social proximity to specific PPP lenders

Read:

Table 7

Social proximity to lenders

Dep. Variable:	FinTech Per Capita	Womply Per Capita		BlueAcorn Per Capita	BoA Per Capita	JPMChase Per Capita	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Social Proximity to FinTech	0.778*** (6.19)						
Social Proximity to Womply		0.960*** (6.45)	0.746** (2.34)		-0.193 (-0.85)		
Social Proximity to BlueAcorn			0.229 (0.75)	1.104*** (7.67)	1.289*** (5.38)		
Social Proximity to Bank of America						-0.0614 (-1.03)	
Social Proximity to JPMorgan Chase							0.105 (1.62)
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	19,367	19,367	19,367	19,367	19,367	19,367	19,367
Num. Counties	2,320	2,320	2,320	2,320	2,320	2,320	2,320
Adjusted R ²	0.884	0.832	0.833	0.861	0.862	0.788	0.698
Adjusted Within R ²	0.751	0.644	0.645	0.720	0.721	0.723	0.569

This table examines the effects of social proximity to FinTech loans and loans originated by different lenders/platforms on borrowers using a FinTech lender or specific lender/platform. Social proximity to lending by a lender or group of lenders is calculated similarly to social proximity to suspicious lending, but the number of flagged loans per capita in each ZIP code is replaced with the number of loans originated by the lender(s) per capita in each ZIP code. The dependent variables are the number of loans originated by the lender(s) per capita in each ZIP code. FinTech lenders are defined as those used by Griffin, Kruger, and Mahajan (2023). Womply collaborated with five lenders to originate loans: Benworth Capital, Capital Plus Financial, DreamSpring, Fountainhead SBF, and Harvest SBF. BlueAcorn partnered with two lenders to originate loans: Capital Plus Financial and Prestamos CDFI. To ensure nationally representative estimates, we use weighted least squares regressions, with weights based on the number of PPP loans in each ZIP code. All variables (both independent and dependent) are standardized to have a mean of 0 and a standard deviation of 1. ZIP codes are filtered to include only those with at least 25 loans. Fixed effects and control variables are specified at the bottom of each column. The controls include the log of population density, percentage nonwhite, log of average household income, poverty rate, prepandemic unemployment rate, percentage with a college education, the shares of Facebook users in the ZIP code with friends living within 50 and 150 miles, and PPP loans per capita. t -statistics based on robust standard errors are clustered by county in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

- Table 7 tests whether socially connected ZIP codes use the same lenders and platforms.
- Social proximity to FinTech lending strongly predicts local FinTech loan use: coefficient **0.778**.
- Social proximity to Womply lending predicts Womply loans per capita: coefficient **0.960**, remaining **0.746** when controlling for BlueAcorn proximity.
- Social proximity to BlueAcorn lending predicts BlueAcorn loans per capita: coefficient **1.104**, remaining **1.289** when controlling for Womply proximity.
- Social proximity to Bank of America or JPMorgan Chase lending is not statistically significant.
- The table supports the view that social networks transmitted specific lender/platform recommendations, especially for FinTech PPP lenders.

6 Short summary

- **Method.** The paper combines loan-level PPP fraud indicators, ZIP-code social connectedness, demographics, social-capital measures, social-media data, and lender-choice data to study whether fraud spreads through social networks.
- **Identification.** The empirical strategy compares ZIP codes within the same county, controls for rich demographics and FinTech exposure, separates social from physical proximity, uses timing across PPP rounds, and examines granular mechanisms such as identical loan features and specific lender choice.
- **Main empirical result.** Social proximity to suspicious lending is the strongest predictor of local PPP fraud. It explains much more variation than demographics, social capital, or crime/corruption proxies.
- **Mechanism.** The evidence is consistent with social networks spreading both knowledge and

normalization of fraud. Social-media groups, repeated loan-feature clusters, and common FinTech lender choices all point toward transmission of specific application methods.

- **Policy result.** Fraud prevention must be faster and more data-driven. Once fraud methods become socially contagious, ex post prosecution alone is too slow to prevent large-scale losses.

7 Conclusion

7.1 Core conclusion

- The paper shows that suspicious PPP lending was not randomly distributed across the United States.
- Fraud was geographically concentrated, grew sharply over time, and was especially pronounced in FinTech lending.
- Social connections strongly predict where fraud occurred and where it subsequently spread.
- The evidence goes beyond aggregate correlations: social connections also predict social-media exposure, identical loan-application features, and specific FinTech lender choices.

7.2 Economic intuition

- Fraud has an information component: potential borrowers need to learn that the opportunity exists, how to apply, which lender to use, what documents to submit, and what amounts or employee counts to report.
- Fraud also has a norms component: people may be more willing to participate if they observe others doing it or if online groups normalize it as low risk or socially acceptable.
- Social networks can transmit both the technical information and the moral rationalization.
- Once this process begins, weak screening and fast disbursement can turn scattered fraud into a nationwide cascade.

7.3 Takeaway

This paper treats fraud not only as individual misconduct but as a contagious financial behavior. In the PPP setting, social networks appear to have transmitted fraud opportunities, instructions, platform recommendations, and normalization. The main lesson is that government programs involving large rapid payments need proactive data analysis and early intervention, because once fraud becomes socially transmissible, “pay now, chase later” can allow losses to scale far beyond what traditional enforcement capacity can recover.